The empirical recurrence for OEIS A269410, proved from an exact model

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Abstract

OEIS A269410 carries an empirical recurrence of order 5. It is true, and it is decidable rather than empirical. The entry's sequence is given exactly by an exact polynomial in n, from which a provably satisfies a MONIC linear recurrence of order $S = 5$, derived below and not assumed. The residual of the conjectured recurrence satisfies the same one, so whether it annihilates a is settled by a finite exact computation. No transfer matrix is involved and the count is not a walk.

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1 The conjecture

OEIS A269410 is "Number of length-4 0..n arrays with no repeated value greater than or equal to the previous repeated value.". It has offset 1 and begins

$$9, 63, 222, 570, 1215, 2289, 3948, 6372, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: a(n) = n^4 + 4*n^3 + (7/2)*n^2 + (1/2)*n.

As of the "Last modified" line on the live entry (Jan 21 2019 10:06:51, revision 7) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a polynomial

The array has a FIXED length L and an alphabet $0..n$ that grows. The condition is decided by the ORDER of the terms alone — by which are equal and which is larger — so an admissible array is described completely by its weak ordering: the ordered set partition of the L positions into m blocks of equal value, blocks listed in increasing order. Every assignment of m distinct values from $\{0, \ldots, n\}$ to those blocks, in increasing order, gives one array, and there are $\binom{n+1}{m}$ of them. Hence

$$a(n) \ = \ \sum_{m=1}^{L} N_m \binom{n+1}{m},$$

with N_m the number of admissible weak orderings into m blocks, computed exactly.

3 The bound

That expression is a polynomial in n of degree at most L, so $(z-1)^{L+1}$ annihilates a and $S = 5$. Nothing is fitted and no threshold is guessed: the N_m are counted, not estimated.

4 The proof

Lemma 1. *Let* $A(z) = z^S - \sum_{j<S} \alpha_j z^j$ *be the monic annihilator derived above and* $q(z) = z^5 - \sum_i c_i z^{5-i}$ *the characteristic polynomial of the conjectured recurrence. The residual* $r(n) = a(n) - \sum_i c_i a(n-i)$ *is a linear combination of shifts of* a*, so* A *annihilates* r *as well. Since* $A(0) \neq 0$ *the recurrence it gives runs in both directions, and* S *consecutive zeros of* r *force* r *to vanish at every later index.*

Proof. A annihilates a, hence every shift of a, hence any linear combination of shifts; that is r. A monic recurrence with nonzero constant term determines each term from the S before it and each from the S after it, so a run of S zeros propagates. □

Theorem 2.

$$a(n) \;=\; 5\,a(n-1) - 10\,a(n-2) + 10\,a(n-3) - 5\,a(n-4) + a(n-5)$$

for every $n > 0$*.*

Proof. The terms $a(0), \ldots$ were computed exactly in integer arithmetic from the model, extended by A where the model itself was not evaluated, and $r(n)$ formed for each index. Those integers vanish from $n = 0 + 1$ onwards, and the run exceeds $S + 5$, which by Lemma 1 settles every larger index at once. □

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 33 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: it is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no model involved, and holds wherever the proved range and the data overlap.

Third, the derived annihilator A was itself tested on terms it had not been given: the model was evaluated beyond the S values that determine it and A reproduced them. A bound that is too small fails this test, which is why it is run.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A269410.

[2] M. Beck and S. Robins, *Computing the Continuous Discretely*, 2nd ed., Springer, 2015. (Ehrhart quasi-polynomials and their periods.)

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011.

[4] F. Diamond and J. Shurman, *A First Course in Modular Forms*, Springer GTM 228, 2005.